CROW CARRIER — eight synchronous microphone channels / USB data / 12 V operating power

PROTOTYPE — NOT FOR FABRICATION. Outdoor protection and exact assembly qualification remain open.

12 V INPUT / REVERSE PROTECTION / 5 V CONVERSION

J1
VIN12 1
GND 2
CENTER_12V
SLEEVE_GND
PJ-102AH / 12V CENTER POSITIVE

F1
VIN12
0455001.MRL 1A
F1608I2

D3
SS34-E3/57T

12V

D1
Littelfuse SMBJ15A
GND

C1
10uF 50V X7R 10%
GND

C2
10uF 50V X7R 10%
GND

U10
VIN
GND
VOUT
TSR 1-2450

5V

D2
VOUT
Diodes Inc 1N5819HW-7-F

PWR1
VIN12
PWR_FLAG

PWR2
12V
PWR_FLAG

PWR3
GND
PWR_FLAG

PWR4
VOUT
PWR_FLAG

ADC 1: CHANNELS 1-4 / PCM1864 / common 3V3

U1
ADC1_CH1_M 1 VIN1M
ADC1_CH2_M 2 VIN2M
ADC1_CH1_P 3 VIN1P
ADC1_CH2_P 4 VIN2P
5 MIC_BIAS
REF1 6 VREF
GND 7 AGND
3V3 8 DVDD
9 XO
10 XI
LDO1 11 LDO
GND 12 DGND
3V3 13 DVDD
3V3 14 DVDD
MCLK_ADC1 15 SCKI
24 LRCK
23 BCK
22 DOUT
21 GPO0
20 GPO1
19 MISO
18 SDA
17 SCL
16 MCLKO
GND
27 VINAM
28 VINAP
29 VINBM
30 VINBP
ADC1_CH4_M
ADC1_CH3_M
ADC1_CH4_P
ADC1_CH3_P
PCM1864ADBTR

U1 LOCAL DECOUPLING — separate capacitor positions at each supply pin

REF1
C11
1uF 50V X7R 10% VREF
GND

LDO1
C12
100nF 10V X7R LDO
GND

3V3
C13
100nF 10V X7R AVDD
GND

3V3
C14
10uF 10V X7R AVDD
GND

3V3
C15
100nF 10V X7R DVDD
GND

3V3
C16
10uF 10V X7R DVDD
GND

LDO1
C17
10uF 10V X7R LDO
GND

3V3
C18
100nF 10V X7R IOVDD
GND

ADC 2: CHANNELS 5-8 / PCM1864 / common 3V3

U2
ADC2_CH1_M 1 VIN1M
ADC2_CH2_M 2 VIN2M
ADC2_CH1_P 3 VIN1P
ADC2_CH2_P 4 VIN2P
5 MIC_BIAS
REF2 6 VREF
GND 7 AGND
3V3 8 DVDD
9 XO
10 XI
LDO2 11 LDO
GND 12 DGND
3V3 13 DVDD
3V3 14 DVDD
MCLK_ADC2 15 SCKI
24 LRCK
23 BCK
22 DOUT
21 GPO0
20 GPO1
19 MISO
18 SDA
17 SCL
16 MCLKO
GND
27 VINAM
28 VINAP
29 VINBM
30 VINBP
ADC2_CH4_M
ADC2_CH3_M
ADC2_CH4_P
ADC2_CH3_P
PCM1864ADBTR

U2 LOCAL DECOUPLING — separate capacitor positions at each supply pin

REF2
C21
1uF 50V X7R 10% VREF
GND

LDO2
C22
100nF 10V X7R LDO
GND

3V3
C23
100nF 10V X7R AVDD
GND

3V3
C24
10uF 10V X7R AVDD
GND

3V3
C25
100nF 10V X7R DVDD
GND

3V3
C26
10uF 10V X7R DVDD
GND

LDO2
C27
10uF 10V X7R LDO
GND

3V3
C28
100nF 10V X7R IOVDD
GND

U3 ASSEMBLY: DEPOPULATE ON-MODULE D1 only: retain R10/R1 VBUS sense. USB DATA ONLY.

U3
SDA 1 GPO0
SCL 2 GPO1
GND 3 GND
TDM 4 GPO2
BCK 5 GPO3
LRCK 6 GPO4
BUZZ_GPIO1 7 GPO5
GND 8 GND
BUZZ_GPIO2 9 GPO6
BUZZ_GPIO3 10 GPO7
BUZZ_GPIO4 11 GPO8
BUZZ_GPIO5 12 GPO9
GND 13 GND
BUZZ_GPIO6 14 GPO10
BUZZ_GPIO7 15 GPO11
BUZZ_GPIO8 16 GPO12
17 GPO13
GND 18 GND
19 GPO14
20 GPO15
GPO24 47
GPO23 46
GND 45
GPO22 44
GPO21 43
GND 42
GPO20 41
GPO19 40
GND 39
GPO18 38
GPO17 37
AGND 36
GPO16 35
ADC_VREF 34
3V3 33
3V3_EN 32
GND 31
VSYS 30
VBUS 29
Pico SC0915 MCOs on-module D1 DEPOPULATED

R1
3V3
4.7k 1%
SDA

R2
3V3
4.7k 1%
SCL

SHARED 12.288 MHz MCLK

X1
3V3 1 OE
GND 2 GND
OUT 3
VDD 4
MCLK_SRC
3V3
LMK6CE01228BCDLFR

C4
3V3
1uF 50V X7R 10% CLOCK
GND

R3
MCLK_SRC
MCLK_FAN
68R 1% 0.1W 25V

R4
MCLK_ADC1
56R 1% 0.1W 25V

R5
MCLK_ADC2
56R 1% 0.1W 25V

POD 1 — differential audio + power + active-low calibration command

J101
P1_A_P contact_1 contact_5 GND
P1_A_M contact_2 contact_6 GND
P1_12V contact_3 contact_7 P1_BUZZ_N
P1_12V contact_4 contact_8 GND
RJHSE-5080

D501
P1_A_P 1 CT
GND 2 GND
P1_A_M 3 CT
5 P1_BUZZ_N
6 NC
SP0504BAHTG

POD 2 — differential audio + power + active-low calibration command

J102
P2_A_P contact_1 contact_5 GND
P2_A_M contact_2 contact_6 GND
P2_12V contact_3 contact_7 P2_BUZZ_N
P2_12V contact_4 contact_8 GND
RJHSE-5080

D502
P2_A_P 1 CT
GND 2 GND
P2_A_M 3 CT
5 P2_BUZZ_N
6 NC
SP0504BAHTG

POD 3 — differential audio + power + active-low calibration command

J103
P3_A_P contact_1 contact_5 GND
P3_A_M contact_2 contact_6 GND
P3_12V contact_3 contact_7 P3_BUZZ_N
P3_12V contact_4 contact_8 GND
RJHSE-5080

D503
P3_A_P 1 CT
GND 2 GND
P3_A_M 3 CT
5 P3_BUZZ_N
6 NC
SP0504BAHTG

POD 4 — differential audio + power + active-low calibration command

J104
P4_A_P contact_1 contact_5 GND
P4_A_M contact_2 contact_6 GND
P4_12V contact_3 contact_7 P4_BUZZ_N
P4_12V contact_4 contact_8 GND
RJHSE-5080

D504
P4_A_P 1 CT
GND 2 GND
P4_A_M 3 CT
5 P4_BUZZ_N
6 NC
SP0504BAHTG

POD 5 — differential audio + power + active-low calibration command

J105
P5_A_P contact_1 contact_5 GND
P5_A_M contact_2 contact_6 GND
P5_12V contact_3 contact_7 P5_BUZZ_N
P5_12V contact_4 contact_8 GND
RJHSE-5080

D505
P5_A_P 1 CT
GND 2 GND
P5_A_M 3 CT
5 P5_BUZZ_N
6 NC
SP0504BAHTG

POD 6 — differential audio + power + active-low calibration command

J106
P6_A_P contact_1 contact_5 GND
P6_A_M contact_2 contact_6 GND
P6_12V contact_3 contact_7 P6_BUZZ_N
P6_12V contact_4 contact_8 GND
RJHSE-5080

D506
P6_A_P 1 CT
GND 2 GND
P6_A_M 3 CT
5 P6_BUZZ_N
6 NC
SP0504BAHTG

POD 7 — differential audio + power + active-low calibration command

J107
P7_A_P contact_1 contact_5 GND
P7_A_M contact_2 contact_6 GND
P7_12V contact_3 contact_7 P7_BUZZ_N
P7_12V contact_4 contact_8 GND
RJHSE-5080

D507
P7_A_P 1 CT
GND 2 GND
P7_A_M 3 CT
5 P7_BUZZ_N
6 NC
SP0504BAHTG

POD 8 — differential audio + power + active-low calibration command

J108
P8_A_P contact_1 contact_5 GND
P8_A_M contact_2 contact_6 GND
P8_12V contact_3 contact_7 P8_BUZZ_N
P8_12V contact_4 contact_8 GND
RJHSE-5080

D508
P8_A_P 1 CT
GND 2 GND
P8_A_M 3 CT
5 P8_BUZZ_N
6 NC
SP0504BAHTG

C102
P1_A_P
2.2uF 25V X7R 20%
P1_CP

R102
P1_CP
2.7k 1%
ADC1_CH1_P

D102
BZT52C3V9-7-F
GND

C301
2.2nF 50V C0G 5%
GND

C103
P1_A_M
2.2uF 25V X7R 20%
P1_CM

R103
P1_CM
2.7k 1%
ADC1_CH1_M

D103
BZT52C3V9-7-F
GND

C401
2.2nF 50V C0G 5%
GND

C104
P2_A_P
2.2uF 25V X7R 20%
P2_CP

R104
P2_CP
2.7k 1%
ADC1_CH2_P

D104
BZT52C3V9-7-F
GND

C302
2.2nF 50V C0G 5%
GND

C105
P2_A_M
2.2uF 25V X7R 20%
P2_CM

R105
P2_CM
2.7k 1%
ADC1_CH2_M

D105
BZT52C3V9-7-F
GND

C402
2.2nF 50V C0G 5%
GND

C106
P3_A_P
2.2uF 25V X7R 20%
P3_CP

R106
P3_CP
2.7k 1%
ADC1_CH3_P

D106
BZT52C3V9-7-F
GND

C303
2.2nF 50V C0G 5%
GND

C107
P3_A_M
2.2uF 25V X7R 20%
P3_CM

R107
P3_CM
2.7k 1%
ADC1_CH3_M

D107
BZT52C3V9-7-F
GND

C403
2.2nF 50V C0G 5%
GND

C108
P4_A_P
2.2uF 25V X7R 20%
P4_CP

R108
P4_CP
2.7k 1%
ADC1_CH4_P

D108
BZT52C3V9-7-F
GND

C304
2.2nF 50V C0G 5%
GND

C109
P4_A_M
2.2uF 25V X7R 20%
P4_CM

R109
P4_CM
2.7k 1%
ADC1_CH4_M

D109
BZT52C3V9-7-F
GND

C404
2.2nF 50V C0G 5%
GND

C110
P5_A_P
2.2uF 25V X7R 20%
P5_CP

R110
P5_CP
2.7k 1%
ADC2_CH1_P

D110
BZT52C3V9-7-F
GND

C305
2.2nF 50V C0G 5%
GND

C111
P5_A_M
2.2uF 25V X7R 20%
P5_CM

R111
P5_CM
2.7k 1%
ADC2_CH1_M

D111
BZT52C3V9-7-F
GND

C405
2.2nF 50V C0G 5%
GND

C112
P6_A_P
2.2uF 25V X7R 20%
P6_CP

R112
P6_CP
2.7k 1%
ADC2_CH2_P

D112
BZT52C3V9-7-F
GND

C306
2.2nF 50V C0G 5%
GND

C113
P6_A_M
2.2uF 25V X7R 20%
P6_CM

R113
P6_CM
2.7k 1%
ADC2_CH2_M

D113
BZT52C3V9-7-F
GND

C406
2.2nF 50V C0G 5%
GND

C114
P7_A_P
2.2uF 25V X7R 20%
P7_CP

R114
P7_CP
2.7k 1%
ADC2_CH3_P

D114
BZT52C3V9-7-F
GND

C307
2.2nF 50V C0G 5%
GND

C115
P7_A_M
2.2uF 25V X7R 20%
P7_CM

R115
P7_CM
2.7k 1%
ADC2_CH3_M

D115
BZT52C3V9-7-F
GND

C407
2.2nF 50V C0G 5%
GND

C116
P8_A_P
2.2uF 25V X7R 20%
P8_CP

R116
P8_CP
2.7k 1%
ADC2_CH4_P

D116
BZT52C3V9-7-F
GND

C308
2.2nF 50V C0G 5%
GND

C117
P8_A_M
2.2uF 25V X7R 20%
P8_CM

R117
P8_CM
2.7k 1%
ADC2_CH4_M

D117
BZT52C3V9-7-F
GND

C408
2.2nF 50V C0G 5%
GND

BUZZ_GPIO1
R301
100k 1%
GND

Q101
P1_BUZZ_N
BUZZ_GPIO1
DMN3023L-7
GND

F101
12V
P1_12V
1206L010/30NR

BUZZ_GPIO2
R302
100k 1%
GND

Q102
P2_BUZZ_N
BUZZ_GPIO2
DMN3023L-7
GND

F102
12V
P2_12V
1206L010/30NR

BUZZ_GPIO3
R303
100k 1%
GND

Q103
P3_BUZZ_N
BUZZ_GPIO3
DMN3023L-7
GND

F103
12V
P3_12V
1206L010/30NR

BUZZ_GPIO4
R304
100k 1%
GND

Q104
P4_BUZZ_N
BUZZ_GPIO4
DMN3023L-7
GND

F104
12V
P4_12V
1206L010/30NR

BUZZ_GPIO5
R305
100k 1%
GND

Q105
P5_BUZZ_N
BUZZ_GPIO5
DMN3023L-7
GND

F105
12V
P5_12V
1206L010/30NR

BUZZ_GPIO6
R306
100k 1%
GND

Q106
P6_BUZZ_N
BUZZ_GPIO6
DMN3023L-7
GND

F106
12V
P6_12V
1206L010/30NR

BUZZ_GPIO7
R307
100k 1%
GND

Q107
P7_BUZZ_N
BUZZ_GPIO7
DMN3023L-7
GND

F107
12V
P7_12V
1206L010/30NR

BUZZ_GPIO8
R308
100k 1%
GND

Q108
P8_BUZZ_N
BUZZ_GPIO8
DMN3023L-7
GND

F108
12V
P8_12V
1206L010/30NR

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